FACULTY OF COMPUTING

SEMESTER II 2023/2024

SECP2613 SYSTEM ANALYSIS AND DESIGN

SECTION 1

PROJECT PROPOSAL AND PLANNING

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1.0 Introduction

In these fast-paced times, an efficient document management system (DMS) has become essential for every organization. An efficient DMS can significantly reduce the time required for document handling. However, some government agencies, such as **Lembaga Kemajuan Pertanian Kemubu (KADA)**, still face challenges with their DMS.

KADA representative Mr. Ahmad Rohailan Bin Hani stated that the agency still relies on the physical method of using physical forms for its document management. This leads to low efficiency and higher time consuming for users. Additionally, the current DMS lacks comprehensive reporting features necessary for conduct decision-making. As a result, the organization struggles to process documents efficiently and adapt to rapid changes in the information storage. The reliance on physical forms also increases the risk of document loss and makes data retrieval cumbersome.

Mr. Ahmad Rohailan Bin Hani aims to enhance the current DMS to accelerate document processing and improve efficiency. Moreover, for the organization's better development, he hopes to integrate features such as a robust reporting system to facilitate decision-making. By implementing these improvements, KADA can streamline operations, reduce manual workload, and provide better services to its member.

2.0 Information Gathering Process

2.1 Method used

Our team, TechAway chose interviewing as the interactive method for the information gathering process.

For the interview preparation, we started by reading background information to have a better understanding of Lembaga Kemajuan Pertanian Kemubu (KADA). This is done by browsing through all the information featured on the official website. Next, we listed our interview objectives to make sure the interview does not divert from the issues that need to be addressed. Then, we decided on whom we are going to interview. Lastly, we prepared both open-ended and close questions for our interview session.

Dr. Iqbal and Dr. Aryati assisted us in arranging an interview session with KADA's representative, Mr. Ahmad Rohailan Bin Hani through the Webex platform on May 14, which took place at the Hyflex Classroom, N28a Level 1, Faculty of Computing. Through this interview, we identified the limitations of the current system and specific requirements of our client to reduce workload. During the Q&A session after the interview, Mr Ahmad cleared our doubts by providing detailed insights regarding the system to be proposed by us.

2.2 Summary from method used

In conclusion, having an interview with our client is important to collect data and information on human and system requirements. Below are examples of the two types of questions types used in the interview.

Example of close-ended questions

1. Does one need to wait until their retirement before they can withdraw their dividend?

   Answer: No

2. Did KADA receive any complaints regarding the current system implemented?

   Answer: No

3. For the current system in use, will the records filled in the forms be uploaded to online storage or the forms will be physically stored?

   Answer: They will be physically stored

Example of open-ended question

4. What details need to be included in the annual report besides the success rate of application?

   Answer: Financial report and report of every activity held that year.

5. What is the main agenda of KADA to upgrade the current system?

   Answer: Upgrade the current system into an online system which enables checking of loan balance and account details.

6. If the members want to check their savings details or annual report, how they can obtain these information?

   Answer: They can contact the office

3.0 Background Study

KADA currently relies on a Document Management System (DMS) that has been in use for the past 15 years. This system requires filling out physical forms to record data. Users must complete a form to apply for membership or a loan and then submit it for processing.

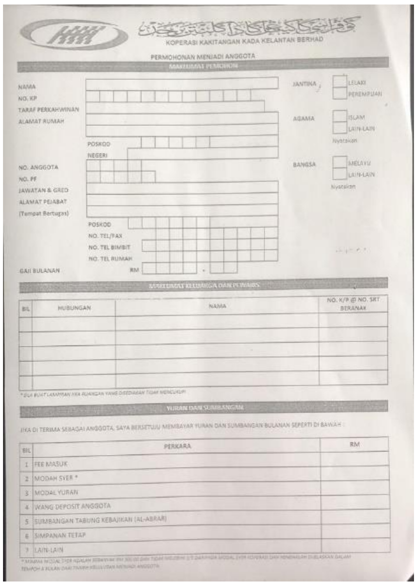


Figure 3.1

Figure 3.1 is a KADA member registration form. KADA current membership registration process relies on a paper form that applicants must fill out and submit. The process requires individuals to provide personal, employment and family information and agree to various fees and contributions. However, this manual system takes a long time, it also prone to data entry errors, and requires a lot of physical storage space. In addition, it limits the accessibility of potential members who cannot easily obtain or submit the form in person.

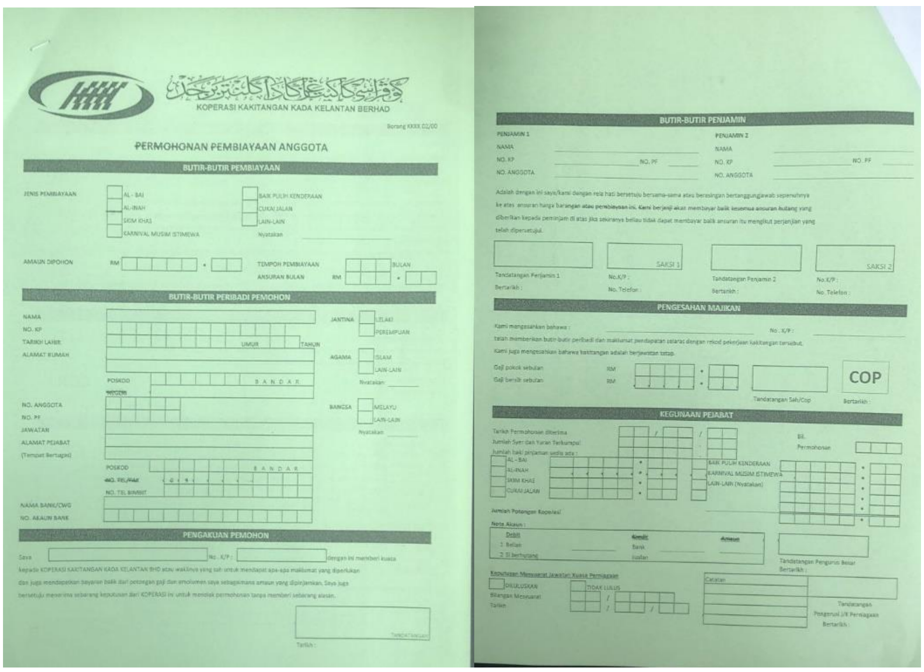
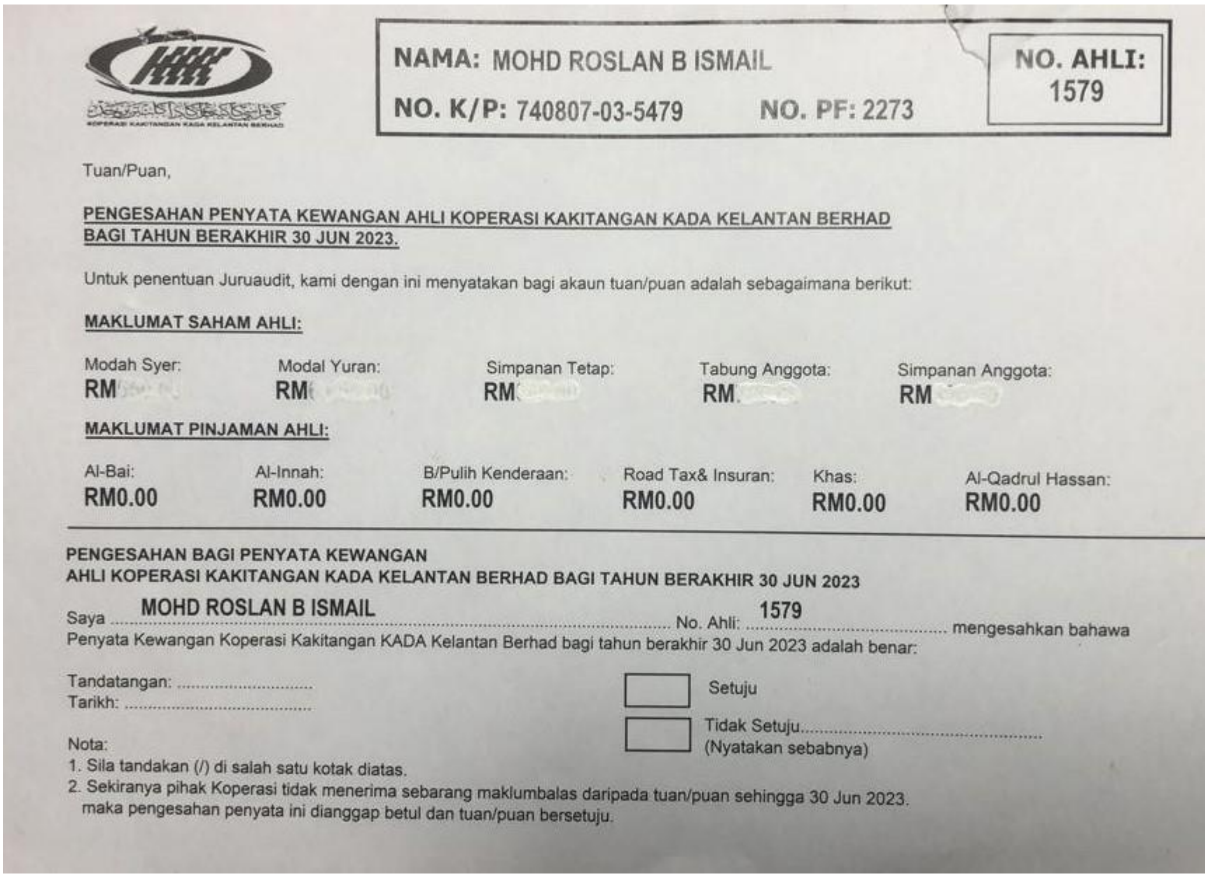


Figure 3.2

Dengan ini memberi kuasa kepada KOPERASI KAKITANGAN KADA KELANTAN BHD atau wakilnya ynag sah untuk mendapat apa-apa maklumat ynag diperlukan dan juga mendapatkan bayaran balik dari potongan gaji dan !$#@ saya sebagaimana amaun yang dipinjamkan. Saya juga bersetuju manerima sebarang keputusan dari KOPERASI ini untuk menolak permohonan tanpa memberi sebarang alasan.

Figure 3.2 show KADA loan application form. A good loan application process should be user-friendly, efficient, accurate, secure, accessible online, and transparent. But KADA current manual process is time-consuming, error-prone and requires physical storage. Digital systems will simplify the applications, improve efficiency, reduce errors, improve accessibility, and secure data by storing it in a data store.



NAMA: MOHD ROSLAN B ISMAIL

NO. K/P: 740807-03-5479 NO. PF: 2273

NO. AHLI: 1579

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modah Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM [illegible]	RM [illegible]	RM [illegible]	RM [illegible]	RM [illegible]

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	B/Pulih Kenderaan:	Road Tax& Insuran:	Khas:	Al-Qadrul Hassan:
RM0.00	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00

PENGESAHAN BAGI PENYATA KEWANGAN
AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023

Saya MOHD ROSLAN B ISMAIL No. Ahli: 1579 mengesahkan bahawa Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar:

Tandatangan: ....
Tarikh: ....

☐ Setuju

☐ Tidak Setuju...
(Nyatakan sebabnya)

Nota:
1. Sila tandakan (/) di salah satu kotak diatas.
2. Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023, maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

Figure 3.3

Figure 3.3 is a KADA member financial statements. As we can see, the current process for generating and distributing member financial statements at KADA involves manually collecting financial data from various records and compiling it into individual statements. These statements include details such as the member's name, identity card number, shares held, loans taken, and savings. This manual system is time-consuming, prone to errors, and inefficient, leading to delays and inaccuracies in financial reporting. By digitizing the process of generating and distributing member financial statements at KADA, it can streamline operations, reduce errors, and improve efficiency. Automation will save time, ensure accuracy, and allow for easier access to up-to-date member information

By looking at the existing system above, we can clearly see that due to manual handling, data transfer and verification take a significant amount of time. So Mr. Ahmad Rohailan aims to enhance the current system by transitioning to an Electronic Document Management System (EDMS), which incorporates electronic elements to manage documents, including the use of spreadsheets and electronic data extraction.

Moreover, the KADA system lacks a comprehensive reporting system. Mr. Ahmad Rohailan has identified this as a major issue. A well-developed reporting system is crucial for decision-making. Currently, KADA faces a shortage of data resources, which hampers its ability to make quick decisions. In today's fast-paced environment, speed is critical for efficiency, as it can secure more opportunities and resources for the organization.

Additionally, members cannot access their account details anytime, anywhere, due to the absence of an online channel. Members must visit the department in person to inquire about their account details or dividend amounts. This increases the workload for agency staff, who must retrieve and convey the account information, leading to long waits and member dissatisfaction.

To address these challenges, KADA intends to upgrade to an EDMS for user registration and the user loan application process. The primary goal of this upgrade is to digitize the document filling process, enhance work efficiency, and strengthen document security. By adopting this innovative and high-tech system, KADA aims to provide a seamless user experience and make informed, timely decisions, ultimately achieving significant organizational improvements.

4.0 Problem Statement

Based on the interview below are the problems that we identified:

Problem 1: Inefficient Manual Data Entry Process

KADA system currently relies on a manual, paper-based system for member registration and loan applications. This process involves several steps: new user or members need to fill out paper forms, and then KADA staff must manually input the collected data into a computer. This method is inherently inefficient, as it doubles the effort required to capture member information. The manual process not only consumes considerable staff time and resources but also leads to potential delays in service delivery. In addition, the reliance on paper forms makes it difficult to maintain a streamlined, centralized, and easily accessible record of member information, further complicating administrative tasks and reporting.

Problem 2: Limited Data Storage Capacity

The existing digital storage capacity for KADA’s member information is severely limited, currently able to accommodate information for only a few hundred members. As KADA aims to expand its membership base, this storage limitation becomes a significant barrier to growth. The current system’s inability to handle a larger volume of data restricts the organization’s potential to attract and manage more members efficiently. This limitation also impacts the organization’s ability to maintain comprehensive records, generate detailed reports, and perform effective data analysis, all of which are critical for strategic planning and operational efficiency.

Problem 3: Security

There are serious security issues with the existing system because it uses paper forms and manual data entry. Paper forms are vulnerable to loss, theft, and unauthorized access, which can compromise sensitive member information. Additionally, the manual data entry process increases the potential of human error. These errors can include inaccurate data entry, misplacement of forms, and delays in processing applications.

Problem 4: Lack of resources to support decision making

KADA's current manual process of transferring information from paper to digital formats is insufficient for effective decision making. This results in fragmented and unstructured data, making it difficult to generate detailed reports and monitor trends. The manual process also limits advanced data analysis, crucial for strategic planning and operational optimization. A system that digitizes member information and organizes data is needed to support informed decision making.

5.0 Proposed Solution

Based on the observation through e-government website, the example that related to the system that we would like to propose include e-Tanah, e-consent, e-Kehakiman, e-Filing and many more. Given the growing trend of processes going paperless, we are going to propose a system to KADA which will serve as an advanced replacement for the traditional paperwork. Below are some features to enhance the efficiency and security of KADA's system.

1. Digitise the member registration and data entry process

   New users or members no longer need to fill out paper forms and KADA staff are not required to perform manual data entry to register new members. This will mitigate the risks of using paper forms, ensure data security and most importantly avoid human error.

2. User authentication mechanism

   After having registered an account, the members can login by entering their username and password. Then, they can apply for a loan or check the status of loan applications and payments made.

3. Automated Generation of Report

   Currently, KADA is still required to generate reports for the board of directors manually. However, by using our system, the report for each member will be automatically generated after the clerk has updated the loan and dividend details in the system. Members can also obtain a copy of their individual report via the homepage.

5.1 Feasibility Study

5.1.1 Technical Feasibility

Our system uses a website as its platform, which only requires users to have a device and internet connection to access the system. However, this system must be equipped with a database system and a server to store the users' information in order to register a new account. Besides, our system will store all the relevant information of our users' savings account information. Our team have sufficient technical resources to provide the level of technology required for this system. Hence, our system is technically feasible.

5.1.2 Operational Feasibility

From the interview with the representatives of KADA, they need this system to reduce the workload of generating reports for users manually. Hence, our system is operationally feasible. However, since this system is connected to a server and several databases, it requires information system (IS) support to prevent bugs and glitches. Our team has the capability of handling this, thus our system is proved to be operationally feasible.

5.1.3 Economic Feasibility

CBA

6.0 Objectives SDLC

In order to develop a proposed system, below are the objectives that need to be considered:

1. Identifying the problems of current system
2. Determine human information requirements of the system to be proposed
3. Analysing system needs after interviewing client
4. Designing the recommended system according to client requirements.

7.0 Scope of the Project

We are committed to developing a system tailored to the needs of the KADA agency. Our system currently focuses on five key modules: user registration, user login, loan application, money savings, and reporting. It is designed exclusively for web use and is not optimized for mobile devices currently. The system is accessible to four types of entities: new users, members, clerk, board of directors.

New Users:

New users engage primarily with the user registration module. They apply for membership by filling out an electronic form, submitting their personal data into the system. After processing, the application results are communicated back to the new users, informing them whether they have successfully become members.

Members:

Members interact with four modules: user login, loan application, money savings, and reporting. Members must log in to authenticate before proceeding with any actions. Once logged in, they can apply for loans by providing the necessary information through an electronic form for senior management review. After processing, the application results are communicated to the members. Additionally, RM50 is automatically deducted from members' monthly salaries for savings. Members can also access various reports within the system to understand institutional interest rates or their returns.

Clerk:

The clerk is involved mainly in the electronic document management of user registration and loan applications. The clerk reviews new user applications to ensure document accuracy and inputs the data back into the system for further review. A similar procedure is followed for loan applications, where the clerk verifies the documents and re-enters the checked information into the system.

Board of Directors:

The Board of Directors primarily reviews the applications processed by the clerks, including membership and loan applications. After discussion, the board members enter their decisions back into the system, and the results are communicated to the applicants. Additionally, the board designs the report module and frequently requires extensive data sources and reports to inform their decisions.

8.0 Project Planning

8.1 Human Resource

We will outline a realistic and feasible human resources plan that aligns with our current skill levels, the need for collaboration, and the support required from experienced mentors and advisors.

Position	Project Coordinator	Business Analyst	System Designer	UI/UX Designer
Lead Member(s)	Gui Kah Sin	Tan Zhi Ming Sabrina Heng	Brendan Chia	Poh Lok Yee
Responsibilities	Organize project activities, ensure timelines met, assisting team members and KADA representatives in communicating.	Gather and analyse requirements from KADA, ensure alignment with KADA's needs.	Design the structure of the digital system	Design user-friendly interfaces for the digital system
Skills Required	Basic project management, good communication and organizational skills.	Analytical thinking, good communication skills	Basic understanding of system design concepts, database concept and user interface design	Basic UI/UX design concepts and familiar with design tools

Figure 8: Workforce distribution of TechAway

<u>Human Resources Plan</u>

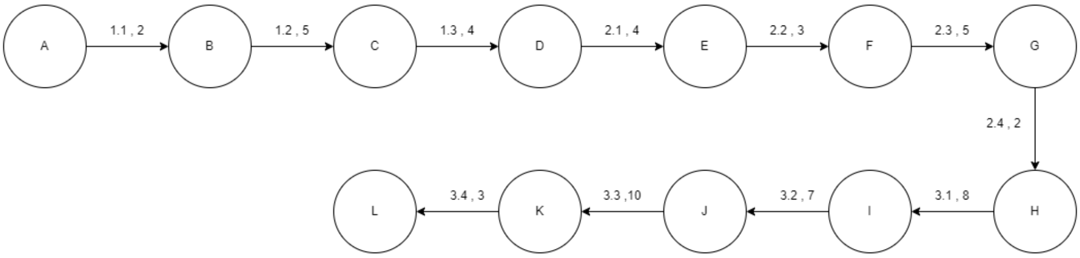
1. Allocation – Assign team roles based on interests and individual strengths within group.
2. Team Development – Organize meeting and training session to train team members with roles, project objectives, and necessary skills.
3. Performance Monitoring and Communication – Plan frequent team check-ins to talk about the status, resolve issues, and get mentor help as needed.

Our human resources plan ensures clear roles and support for each team member, enabling us to effectively manage and execute KADA's digital transformation project.

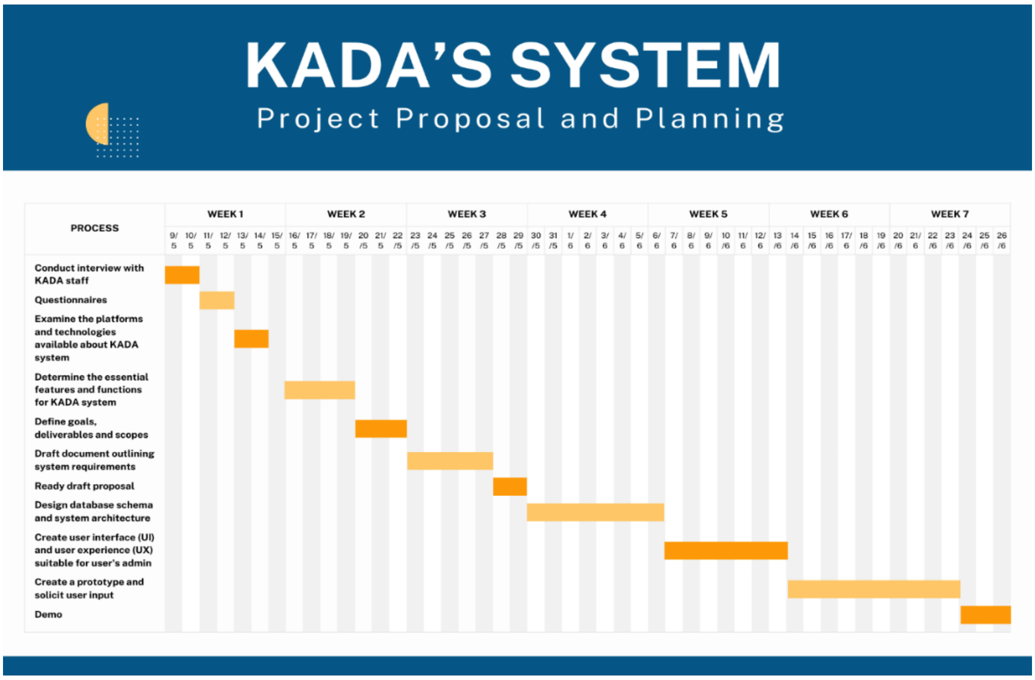
8.2 WBS

No	Activity	Predecessor	Duration (days)
1.0	Research Phase	-	7
1.1	Conduct interview with KADA staff	None	2
1.2	Questionnaires	1.1	5
1.3	Examine the platforms and technologies available about KADA system	1.2	4
2.0	Planning Phase	Research Phase	14
2.1	Determine the essential features and functions for KADA system	1.3	4
2.2	Define goals, deliverables and scopes	2.1	3
2.3	Draft document outlining system requirements	2.2	5
2.4	Ready draft proposal	2.3	2
3.0	Design Phase	Planning Phase	28
3.1	Design database schema and system architecture	2.4	8
3.2	Create user interface (UI) and user experience (UX) suitable for user's admin	3.1	7
3.3	Create a prototype and solicit user input	3.2	10
3.4	Demo	3.3	3

8.3 PERT Chart (based on WBS)



8.4 Gantt Chart

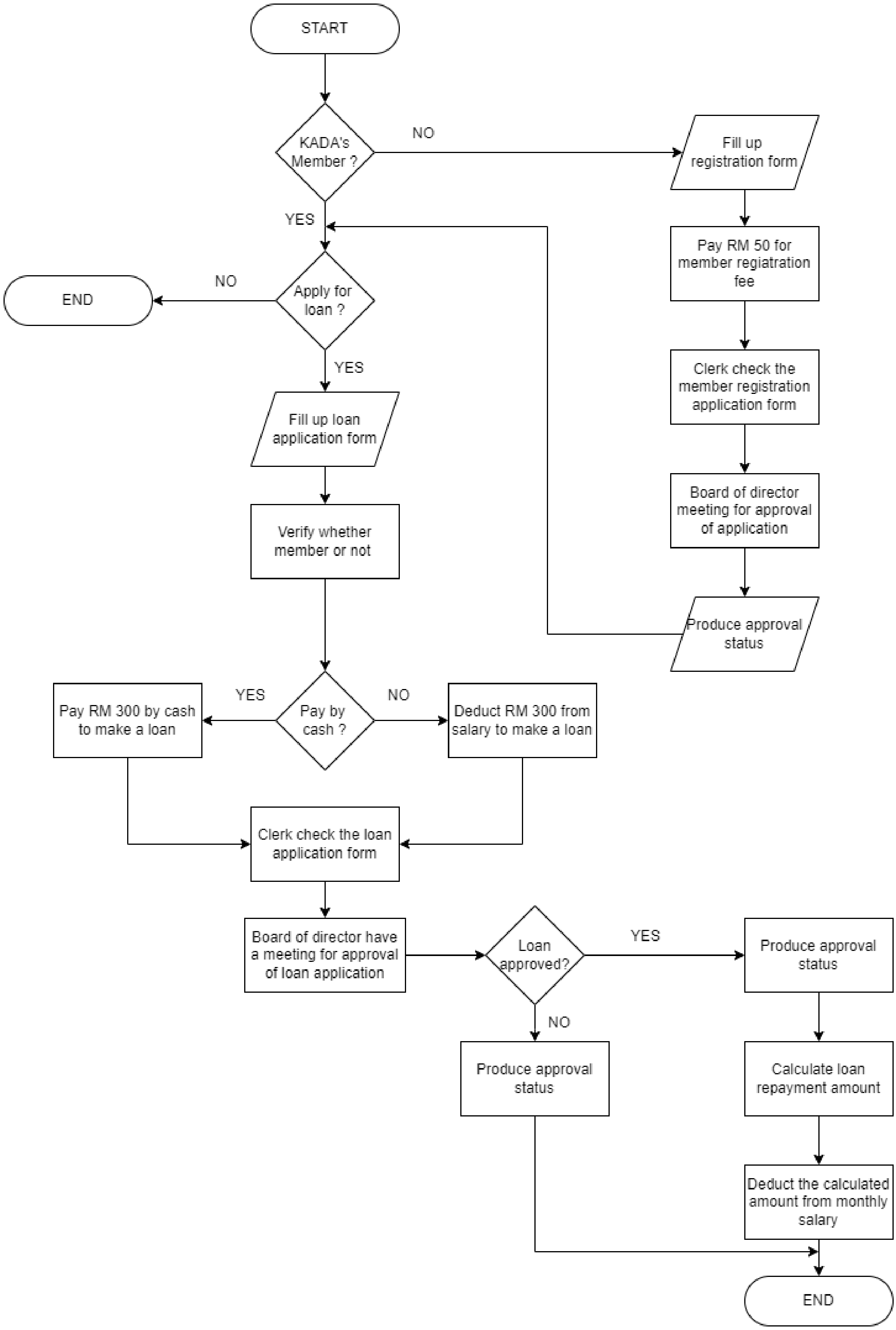


9.0 Requirement Analysis

9.1 Current business process (scenarios, workflow)

In the existing system, employees interested in applying for membership or a loan must individually complete application forms and submit them to the clerk for information verification. Subsequently, the clerk reviews all application details and forwards the validated forms to the board of directors. After that, the board of director will have a meeting for the approval for loan and member application. Next, after meeting the clerk will inform the approval status to the applicant.

Diagram below shows the flowchart for the current business process:


START
KADA's Member ?
NO
Fill up registration form
Pay RM 50 for member regiatration fee
Clerk check the member registration application form
Board of director meeting for approval of application
Produce approval status
YES
Apply for loan ?
NO
END
YES
Fill up loan application form
Verify whether member or not
Pay by cash ?
YES
Pay RM 300 by cash to make a loan
NO
Deduct RM 300 from salary to make a loan
Clerk check the loan application form
Board of director have a meeting for approval of loan application
Loan approved?
YES
Produce approval status
Calculate loan repayment amount
Deduct the calculated amount from monthly salary
NO
Produce approval status
END

9.2 Functional Requirement (input, process and output)

The functional requirements for the system are defined through a detailed analysis of the inputs, processes, and outputs for each type of user within the system. This section outlines the specific functions that new users, members, clerks, and the board of directors will perform, highlighting the necessary data inputs, the processes applied to these inputs, and the expected outputs.

For a new user, the input is their user information. This information is processed to register the member, resulting in an output of the member registration status. Members interact with the system by providing inputs such as loan applications and monthly salary deductions. The loan application is processed to apply for a loan, generating an output of the loan application status. The monthly salary deduction is processed into savings, and the output is an individual report.

Clerks handle the registration and loan applications. They receive inputs in the form of new user member registrations and member loan applications. The process involves checking the registrations and loan applications, which results in outputs of filtered member registrations and filtered loan applications, respectively.

The board of directors works with filtered member registrations and filtered loan applications as inputs. They process these to approve registrations and loans, resulting in outputs of member registration status and loan application approval.

This structure ensures that the system is well-organized and efficient, with clear inputs, processes, and outputs for each user type, thereby enhancing the overall functionality and usability of the system.

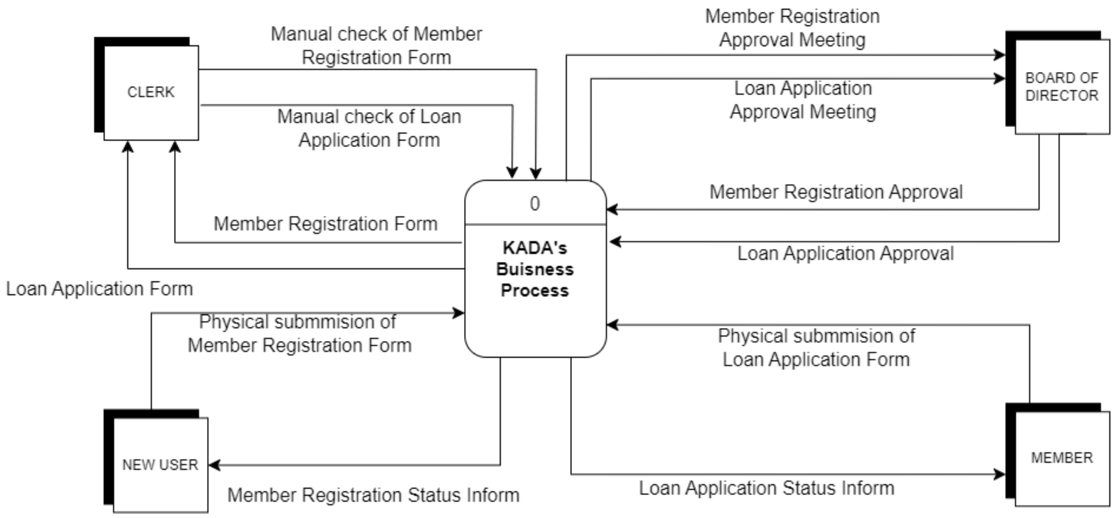
9.3 Non-functional Requirement (performance and control)

- The application forms should be intuitive and easy to navigate
- Verify that the format of user inputs adheres to the required specifications and standards
- Submission of member and loan applications shall be processed and responded to within a short period of time
- The website would instantly update the most current information and data available if any changes occurred
- Create a clear, logical layout with well-labeled sections and buttons, ensuring menus and navigation bars are easy to find
- Regularly create data backups to prevent loss and enable restoration in case of corruption or deletion
- Various payment methods are offered
- Calculation of loan and dividend automatically
- Precise generation of various reports, respectively

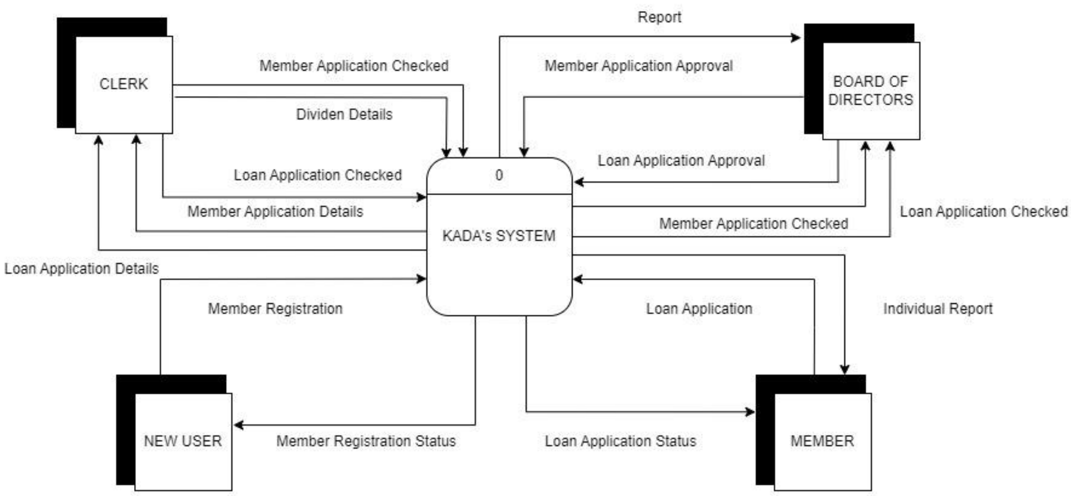
9.4 Logical DFD AS-IS system

9.4.1 Context Diagram

- As – Is

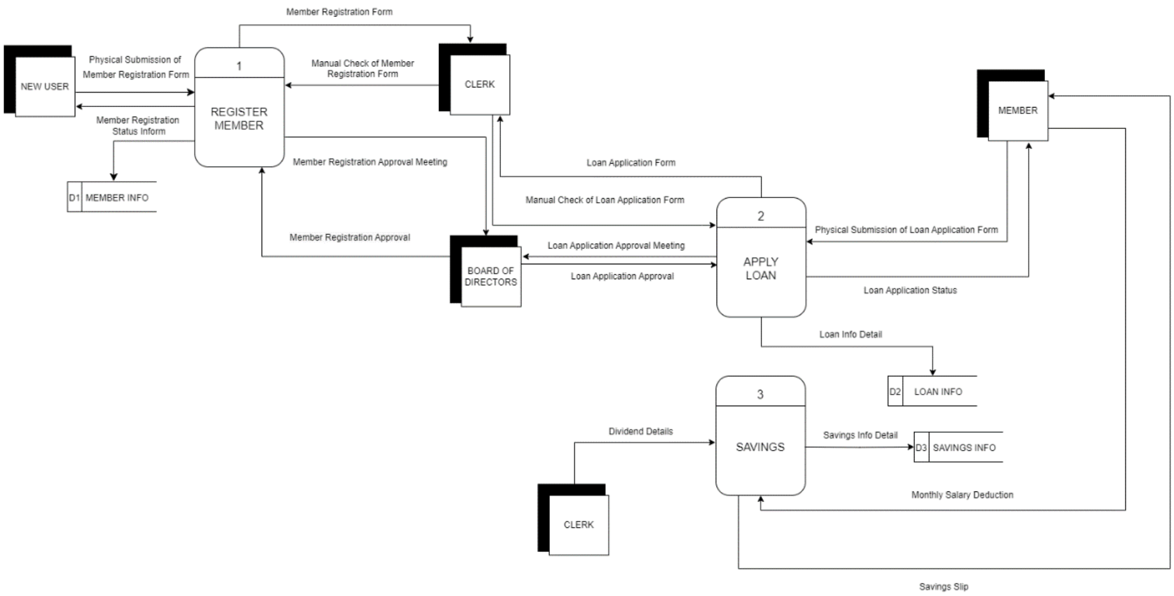


- To Be

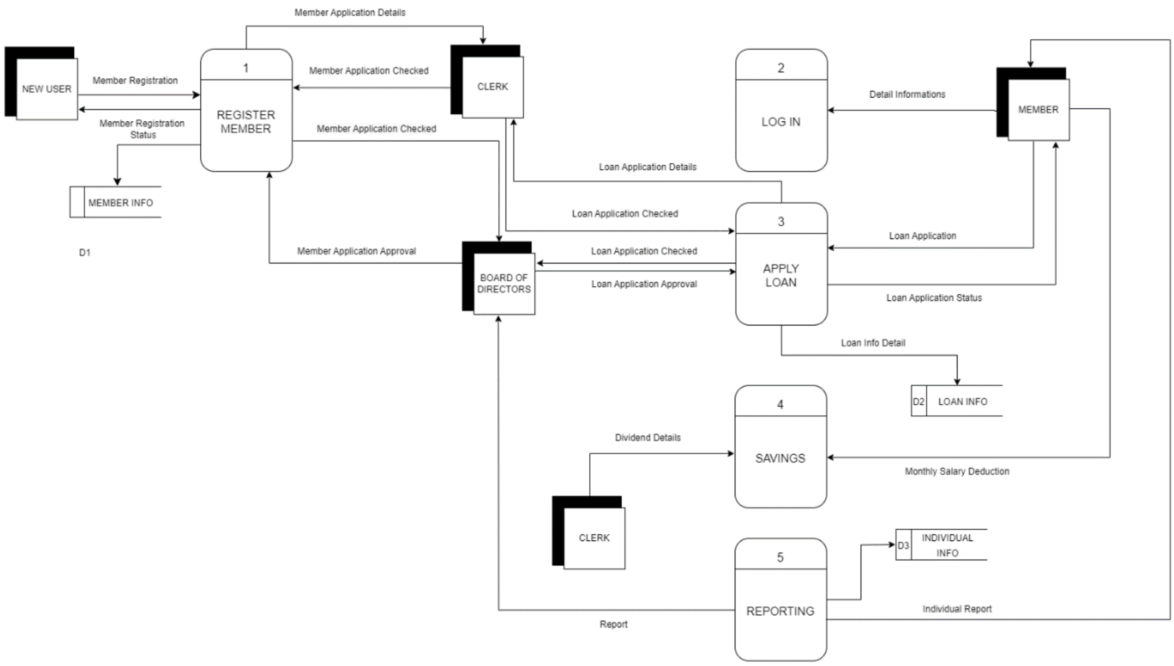


9.4.2 Diagram 0

- As – Is

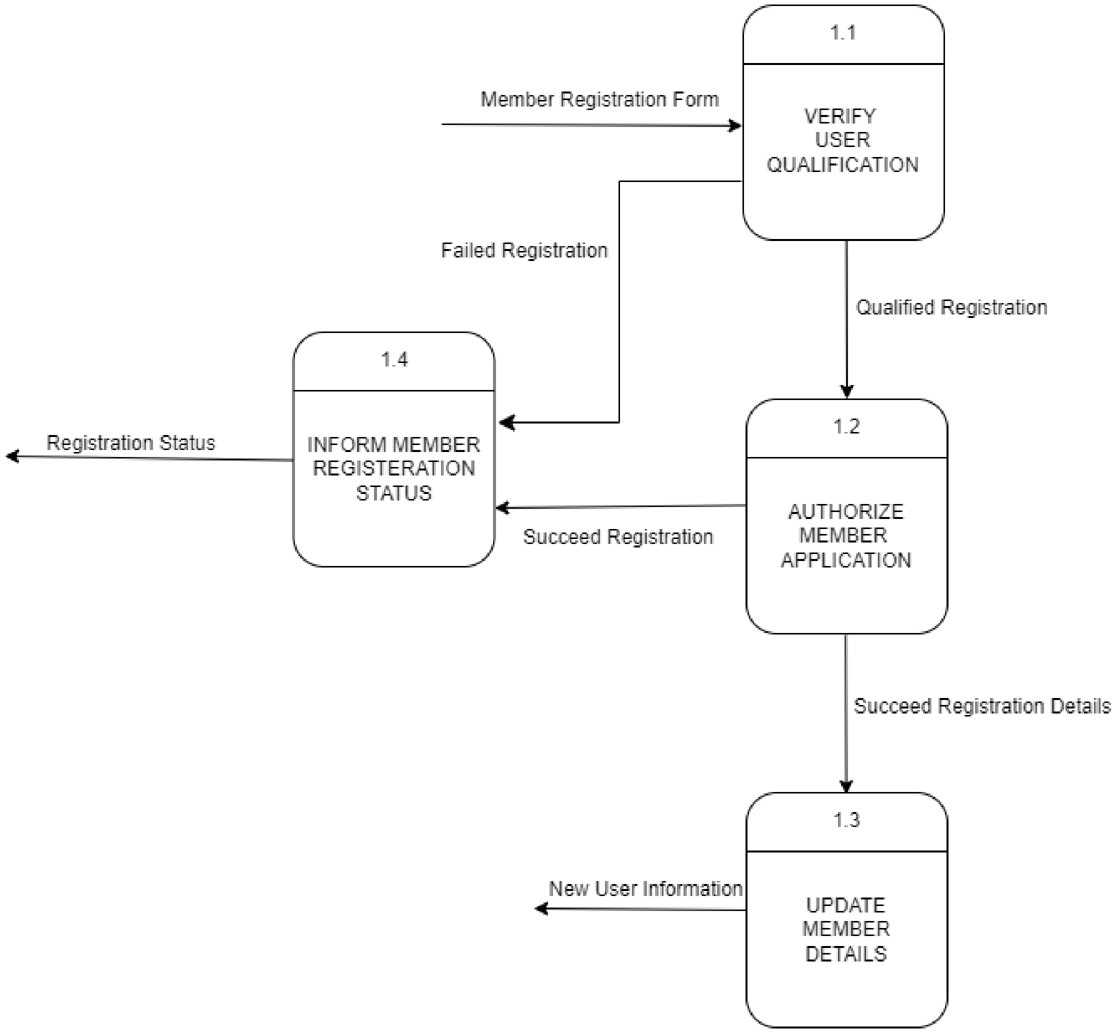


- To Be

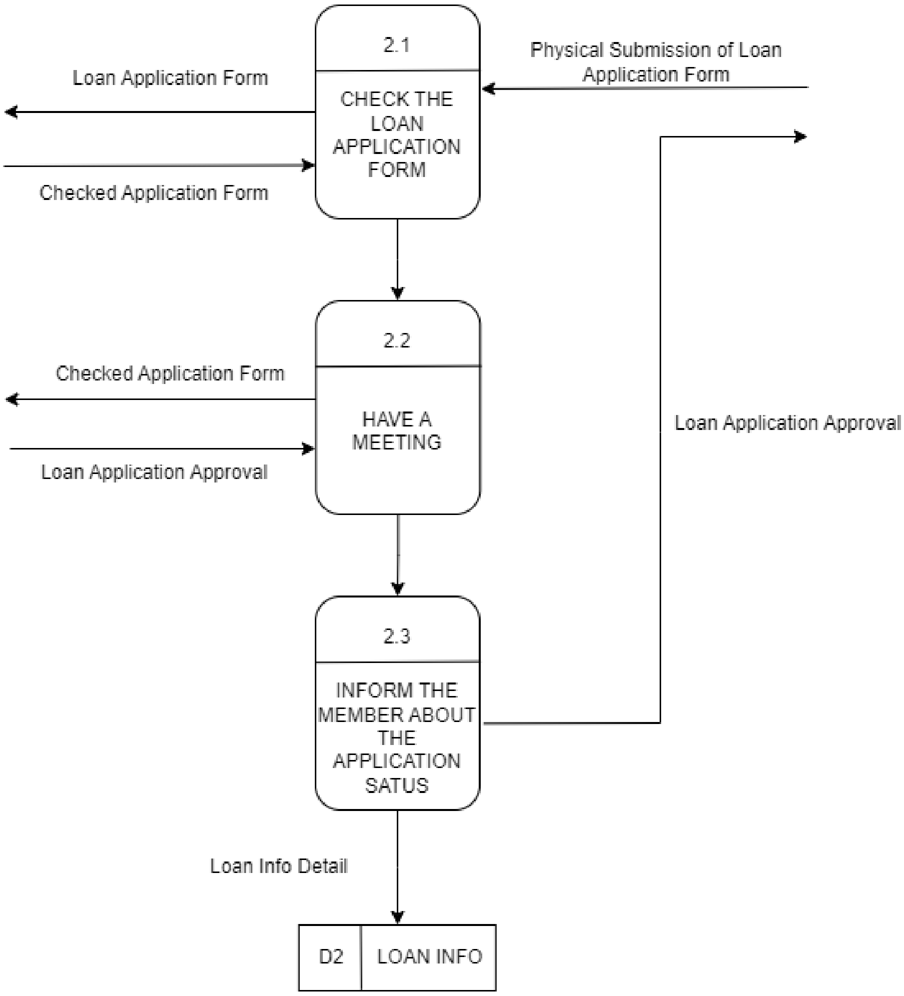


9.4.3 Child Diagram

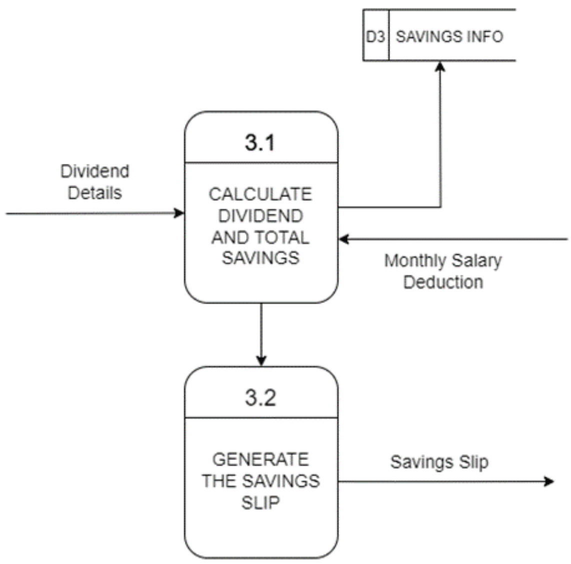
- Process 1 – Register Member



- Process 2 – Apply Loan



- Process 3 – Savings



10.0 Benefit and Overall Summary of Proposed System

1. Enhance Operational Efficiency - Transform the member registration and loan application processes from a manual, paper-based system to a digital one, reducing paperwork and human error.
2. Expand Data Storage Capacity - Upgrade the data storage infrastructure to support a bigger membership base, enabling KADA to manage a growing volume of member information effectively.
3. Facilitate Staff Transition and usage - Provide full instruction and continuing assistance to KADA staff so that they can use the new digital tools and system effectively
4. Support Future Growth – Develop flexible solutions that can adapt to KADA's future needs, ensuring long-term sustainability and operational success.

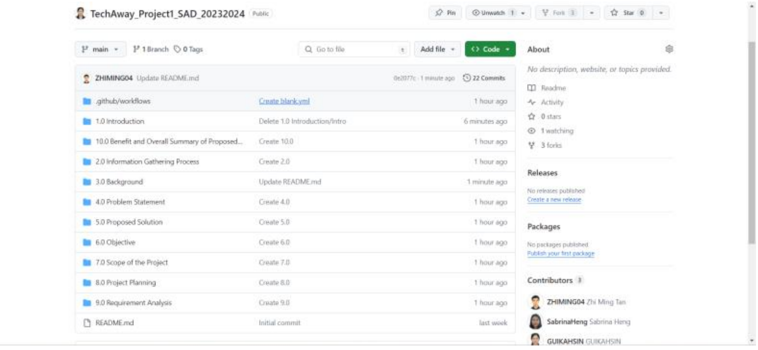
GITHUB

URL of the GitHub Repository

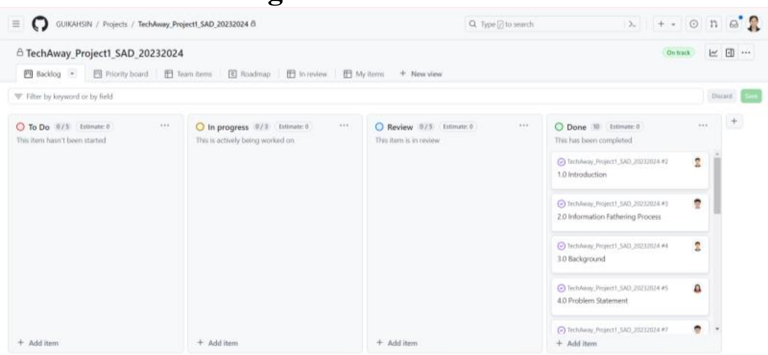
Repository: https://github.com/GUIKAHSIN/TechAway_Project1_SAD_20232024

Project: https://github.com/users/GUIKAHSIN/projects/1

Repository Snapshot



Kanban Board Integration



Version Control Practices

Implementing version control best practices such as feature branching, pull requests, and code reviews significantly enhances team productivity and code quality. Feature branching allows for clear work allocation, ensuring tasks are well-defined and assigned, minimizing overlap and conflicts. Pull requests facilitate structured, timely completion by setting milestones and deadlines, while also incorporating automated checks to quickly identify issues. Code reviews foster comprehensive awareness among team members, promoting shared understanding and knowledge sharing. Continuous integration tools further help in identifying incomplete or faulty code early, ensuring only quality code is merged, thus maintaining an efficient and cohesive development process.